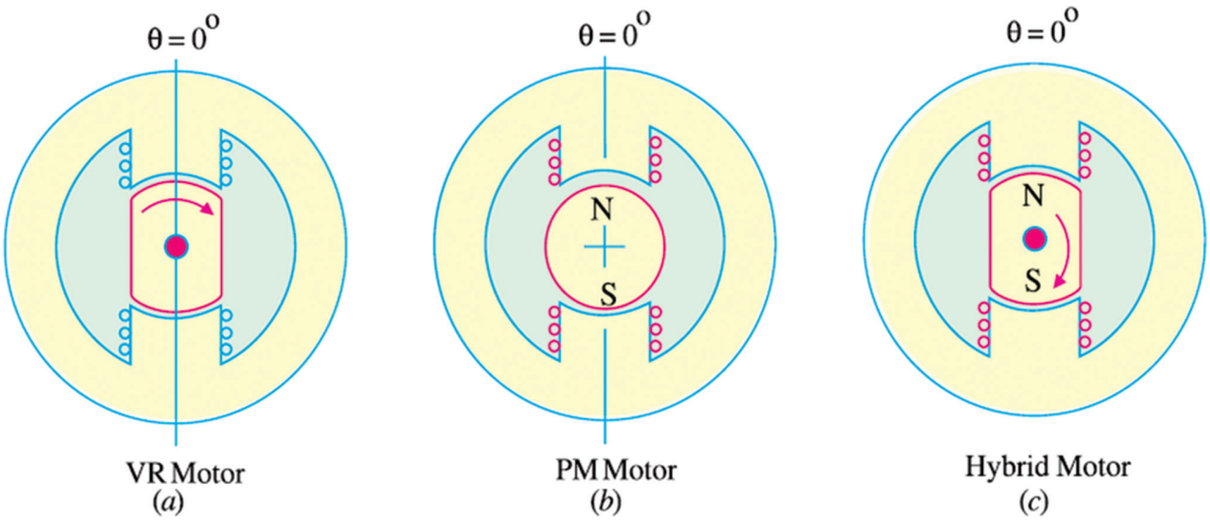


Fig 8.12 *Types of Stepper Motor*

8.6.4 Hybrid Stepper Motor

It combines the features of variable reluctance and permanent magnet stepper motors. This is the most popular type of stepper motor. It has wound stator pole and permanently magnetized rotor poles.

The important features of the hybrid motors are its rotor structure. A recent type motor, a disc rotor is used which is magnetically, axially to give a small stepping angle and low inertia.

Advantages of Stepper Motor

1. The rotating angle of the motor is proportional to the input pulse
2. The motor has full torque at stand
3. Excellent response to starting, stopping and still reversing

Disadvantages of Stepper Motor

1. Resonances can occur if not properly controlled
2. Not easy to operate at extremely high speeds

Uses of Stepper Motor

1. It can be held at a particular position of the shaft
2. Ideal for many autonomous robots requiring higher precision

Points to Remember

1. A.C. generator is also called as alternator or A.C Synchronous generator.
2. Stationary armature and rotating field is most advantageous one.
3. Stator is the stationary part and rotor is the rotating part.
4. Alternator are constructed in two types
   a) Salient pole alternator
   b) Non- salient pole alternator
5. Salient pole type the diameter of stator is large and length is short
6. Non – Salient pole type the diameter of stator is short and length is long
7. A.C motors are classified into
   a) Synchronous motor
   b) Induction motor
8. Synchronous motor is running at synchronous speed which is $N_S = 1500$
9. Synchronous motors are used for power factor correction
10. Induction motor can also be treated as rotating transformer

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